

לציון 7

קורס: ביזיקה קוונטית 1

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פרק: וקטור וסל

$$1 \gg S_{ke}$$

$$\lambda > 0, \quad H = \frac{p^2}{2m} - \lambda \delta(x)$$

למצוא את E האנרגיה והפונקציה $\psi(x)$ הנורמלית

$$E < 0 \quad \text{האנרגיה של המצב הקשור}$$

המצב הקשור הוא $|\psi\rangle$

$$\langle x | H | \psi \rangle = E \langle x | \psi \rangle$$

נציב את $\psi(x)$ במקום $|\psi\rangle$ ונשתמש ב- $\langle x |$

$$-\frac{\hbar^2}{2m} \partial_x^2 \psi(x) - \lambda \delta(x) \psi(x) = E \psi(x)$$

$\Downarrow$

$$\frac{\hbar^2}{2m} \partial_x^2 \psi(x) = (-\lambda \delta(x) - E) \psi(x)$$

ל- $x > 0$ ו- $x < 0$ אין פונקציית דלתא, ולכן:

המשוואה היא $\frac{\hbar^2}{2m} \partial_x^2 \psi(x) = -E \psi(x)$

$$\partial_x^2 \psi(x) = -\frac{2mE}{\hbar^2} \psi(x)$$

$\Downarrow$

||
מכאן נובע

$$\psi(x) = \begin{cases} L_1 e^{\sqrt{-\frac{2mE}{\hbar^2}} x} + L_2 e^{-\sqrt{-\frac{2mE}{\hbar^2}} x} & , x < 0 \\ R_1 e^{\sqrt{-\frac{2mE}{\hbar^2}} x} + R_2 e^{-\sqrt{-\frac{2mE}{\hbar^2}} x} & , x > 0 \end{cases}$$

"גודל" של הנגזרות של פונקציה:

נדרש צדקה $0 < \psi$ פונקציה הולכת $x \rightarrow \pm\infty$

לכן נדרש (כדי שהפונקציה תהיה) $L_2 = R_1 = 0$

נדרש צדקה של פונקציה הולכת $x=0$:

$$L_1 + L_2 = R_1 + R_2$$

||
↓

$$L_1 = R_2$$

נסיים:

$$A = L_1 = R_2$$

דבריו באים צוות: נדרש, הנגזרת של פונקציה

נדרש, כמות - נדרש עבר $0 < \psi$ הולכת:

$$-\frac{\hbar^2}{2m} \int_{-\xi}^{\xi} \partial_x^2 \psi(x) dx - \lambda \int_{-\xi}^{\xi} \delta(x) \psi(x) dx =$$

$$= \left[\int_{-\xi}^{\xi} \psi(x) dx \right] \quad \Downarrow \quad \begin{array}{l} \text{C22N} \\ \text{'3121} \\ \text{E13N: } \mathbb{R} \end{array}$$

$$\lim_{\xi \rightarrow 0} \left[-\frac{\hbar^2}{2m} (\partial_x \psi(\xi) - \partial_x \psi(-\xi)) \right] -$$

$$- \lambda \int_{-\xi}^{\xi} \delta(x) \psi(x) dx = \left[\int_{-\xi}^{\xi} \psi(x) dx \right] \quad \Downarrow \quad \text{S. 1.1.1}$$

$$\lim_{\xi \rightarrow 0} \left[-\frac{\hbar^2}{2m} (\partial_x \psi(\xi) - \partial_x \psi(-\xi)) \right] -$$

$$- \lim_{\xi \rightarrow 0} [\lambda \psi(x)] = \left[\int_{-\xi}^{\xi} \psi(x) dx \right] \quad \Downarrow$$

$\lim_{\epsilon \rightarrow 0} \left[-\frac{\hbar^2}{2m} (\partial_x \psi(\epsilon) - \partial_x \psi(-\epsilon)) \right] -$
 $\lim_{\epsilon \rightarrow 0} [\lambda \psi(0)] = \lim_{\epsilon \rightarrow 0} [2\epsilon \psi(0)]$

$\psi(x) \sim \begin{cases} e^{-\kappa x} & x > 0 \\ e^{\kappa x} & x < 0 \end{cases}$

$$\psi'(0^+) - \psi'(0^-) = \frac{2m\lambda}{\hbar^2} \psi(0)$$

: 2 31

$$\psi(x) = \begin{cases} A e^{\sqrt{-\frac{2mE}{\hbar^2}} x} & , x < 0 \\ A e^{-\sqrt{-\frac{2mE}{\hbar^2}} x} & , x > 0 \end{cases}$$

: 2 31

$$-\sqrt{-\frac{2mE}{\hbar^2}} A e^0 - \sqrt{-\frac{2mE}{\hbar^2}} A e^0 = \frac{2m\lambda}{\hbar^2} A e^0$$

$\Downarrow$

$$-2 \sqrt{-\frac{2mE}{\hbar^2}} = \frac{2m\lambda}{\hbar^2}$$

$$C)^2 \checkmark$$

$$-4 \frac{2mE}{\hbar^2} = \frac{4m^2\lambda^2}{\hbar^4}$$

$$\checkmark$$

$$E = -\frac{m\lambda^2}{2\hbar^2}$$

הנורמה של הפונקציה הנמצאת

הנורמה של הפונקציה הנמצאת

$$\int_{-\infty}^{\infty} |\psi(x)|^2 dx = 2 \int_0^{\infty} |\psi(x)_{x>0}|^2 dx =$$

$\downarrow$
 $\psi(x)$
הנורמה

$$= 2 \int_0^{\infty} |A|^2 e^{-2 \int_0^x \sqrt{-\frac{2mE}{\hbar^2}} x} dx = \frac{2A^2}{-2 \int_0^x \sqrt{-\frac{2mE}{\hbar^2}}} e^{-2 \int_0^x \sqrt{-\frac{2mE}{\hbar^2}} x} \Big|_0^{\infty} =$$

$$= 0 - \frac{2A^2}{-2\sqrt{-\frac{2mE}{\hbar^2}}} = \frac{A^2}{\sqrt{-\frac{2mE}{\hbar^2}}} = 1$$

$\Downarrow$

$$A^2 = \sqrt{-\frac{2mE}{\hbar^2}}$$

$$E = -\frac{m\lambda^2}{2\hbar^2} \quad ; \text{נציב} \quad \Downarrow$$

$$A^2 = \sqrt{\frac{2m \cdot \frac{m\lambda^2}{2\hbar^2}}{\hbar^2}}$$

$$\Downarrow$$

$$A^2 = \sqrt{\frac{m^2 \lambda^2}{\hbar^4}}$$

$$\Downarrow$$

$$A = \sqrt{\frac{m\lambda}{\hbar^2}}$$

כאן הפונקציה — הפונקציה — היא:

$$\psi(x) = \begin{cases} \sqrt{\frac{m\lambda}{\hbar^2}} e^{\frac{m\lambda}{\hbar^2} x} & , x < 0 \\ \sqrt{\frac{m\lambda}{\hbar^2}} e^{-\frac{m\lambda}{\hbar^2} x} & , x > 0 \end{cases}$$

2. 3re

$$\psi(x) = \begin{cases} A \sin^2\left(\pi \frac{(L-x)}{L}\right) & 0 \leq x \leq L \\ 0 & \text{sonst} \end{cases}$$

(1)

$$\int_{-\infty}^{\infty} |\psi(x)|^2 dx = \int_0^L A^2 \sin^4\left(\pi \frac{(L-x)}{L}\right) dx =$$

$$= A^2 \int_0^L \left(\frac{1}{2} - \frac{1}{2} \cos^2\left(2\pi \frac{(L-x)}{L}\right) \right)^2 dx =$$

$$= A^2 \int_0^L dx \left(\frac{1}{4} - \frac{1}{2} \cos^2\left(2\pi \frac{(L-x)}{L}\right) + \frac{1}{4} \cos^4\left(2\pi \frac{(L-x)}{L}\right) \right)$$

$$= A^2 \int_0^L dx \left(\frac{1}{4} - \frac{1}{2} \left(\frac{1}{2} + \cos\left(4\pi \frac{(L-x)}{L}\right) \right) + \frac{1}{4} \left(\frac{1}{2} + \cos^2\left(2\pi \frac{(L-x)}{L}\right) \right) \right)$$

$$= A^2 \int_0^L dx \left(-\frac{1}{2} \cos\left(4\pi \frac{(L-x)}{L}\right) + \frac{1}{8} + \frac{1}{4} \left(\frac{1}{2} + \frac{1}{2} + \cos\left(4\pi \frac{(L-x)}{L}\right) \right) \right)$$

$$= A^2 \int_0^L dx \left(\frac{3}{8} - \frac{1}{4} \cos\left(4\pi \frac{(L-x)}{L}\right) \right) = \frac{3}{8} A^2 L = 1 \Rightarrow$$

$$A = \sqrt{\frac{8}{3L}}$$

(2) סדרת פונקציות

רצפים אינסופיים
קבוצות
האנליזה

$$\psi_E(x) = \sum_{n=1}^{\infty} \langle x | \varphi_n \rangle \langle \varphi_n | \psi \rangle =$$

$$= \sum_{n=1}^{\infty} \varphi_n(x) \int_{-\infty}^{\infty} \varphi_n^*(x) \psi(x) dx \quad (*)$$

↓
רצפים
רצפים
קבוצות
האנליזה

מרחב פונקציות רצפות (כל פונקציה רצפה היא רצפה)

$$\varphi_n(x) = A \cos\left(\frac{\sqrt{2mE_n}}{\hbar} x\right) + B \sin\left(\frac{\sqrt{2mE_n}}{\hbar} x\right)$$

↓
קבוצת פונקציות רצפות

$$\varphi_n(0) = \varphi_n(L) = 0$$

$$A = 0$$

$$B \sin\left(\frac{\sqrt{2mE_n}}{\hbar} L\right) = 0$$

↓

$$\frac{\sqrt{2mE_n}}{\hbar} L = \pi k, \quad k \in \mathbb{Z}$$

↓

$$E_k = \frac{\pi^2 k^2 \hbar^2}{2mL^2}$$

$$\varphi_k(x) = B \sin\left(\frac{\sqrt{2m} \pi k \hbar}{\sqrt{2m} L \hbar} x\right) = B \sin\left(\frac{\pi k}{L} x\right) \quad ; \text{נורמליזציה}$$

נניח B מסתבר על 0 עד 1 :

$$\int_{-\infty}^{\infty} |\varphi_k(x)|^2 dx = |B|^2 \int_0^L \sin^2\left(\frac{\pi k}{L} x\right) dx =$$

נניח B מסתבר על 0 עד 1 :

$$= |B|^2 \int_0^L \left(\frac{1}{2} - \frac{1}{2} \cos\left(\frac{2\pi k}{L} x\right) \right) dx =$$

$$= B^2 \frac{L}{2} = 1 \Rightarrow B = \sqrt{\frac{2}{L}}$$

נניח B מסתבר על 0 עד 1 :

$$(**) \varphi_k(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{\pi k}{L} x\right)$$

נניח B מסתבר על 0 עד 1 :

$$\psi_E(x) = \sum_{i=1}^{\infty} \varphi_i(x) \int_{-\infty}^{\infty} \varphi_i^*(x) \psi(x) dx$$

נניח B מסתבר על 0 עד 1 :

$$\begin{aligned}
 \psi_E(x) &= \sum_{k=1}^{\infty} \sqrt{\frac{2}{L}} \sin\left(\frac{\pi k}{L} x\right) \int_0^L dx \sqrt{\frac{2}{L}} \sin\left(\frac{\pi k}{L} x\right) \sqrt{\frac{8}{3L}} \sin^2\left(\pi \frac{(L-x)}{L}\right) \\
 &= \sqrt{\frac{32}{3L^3}} \sum_{k=1}^{\infty} \sin\left(\frac{\pi k}{L} x\right) \int_0^L dx \sin\left(\frac{\pi k}{L} x\right) \sin^2\left(\pi \frac{(L-x)}{L}\right) = \\
 &= \sqrt{\frac{32}{3L^3}} \sum_{k=1}^{\infty} \sin\left(\frac{\pi k}{L} x\right) \int_0^L dx \frac{1}{2} \left(\sin\left(\frac{\pi k x}{L}\right) - \sin\left(\frac{\pi k x}{L}\right) \cos\left(\frac{2\pi x}{L}\right) \right)
 \end{aligned}$$

$\downarrow$
 ~ 1.5
 ~ 1.5
 ~ 1.5

$$= \sqrt{\frac{32}{3L^3}} \sum_{k=1}^{\infty} \sin\left(\frac{\pi k}{L} x\right) \int_0^L dx \left[\frac{L}{2\pi} (1 - \cos(\pi k)) \left(\frac{1}{k} - \frac{k}{k^2 - 4} \right) \right] =$$

$$\frac{1}{2} \int_0^L \sin\left(\frac{\pi k x}{L}\right) dx = \frac{L}{2\pi k} (1 - \cos(\pi k))$$

$$\text{pro } \sin \alpha \cos \beta = \frac{1}{2} [\sin(\alpha + \beta) + \sin(\alpha - \beta)] : \sim 1.5 : \text{mod}$$

$$\frac{1}{4} \int_0^L \left[\sin\left(\frac{\pi(k+2)x}{L}\right) + \sin\left(\frac{\pi(k-2)x}{L}\right) \right] dx =$$

$$= \frac{L}{4\pi} \left[\frac{1 - \cos(\pi(k+2))}{k+2} + \frac{1 - \cos(\pi(k-2))}{k-2} \right] =$$

$$\begin{aligned}
 &\overset{\cos(x+2\pi)}{=} \overset{\cos(x)}{=} \frac{L}{4\pi} (1 - \cos(\pi k)) \left(\frac{1}{k+2} + \frac{1}{k-2} \right) \overset{\text{new new}}{=} \frac{L}{2\pi} (1 - \cos(\pi k)) \frac{k}{k^2 - 4}
 \end{aligned}$$

נכנס: ψ_1

$$\psi_E(x) = \int \frac{32}{3L^3} \sum_{k=1}^{\infty} \sin\left(\frac{\pi k}{L} x\right) \int_0^L dx \frac{2L}{\pi} \frac{(\cos(\pi k) - 1)}{k(k^2 - 4)} =$$

הנכנס

$$= \int \frac{128}{3\pi^2 L} \sum_{k=1}^{\infty} \frac{(-1)^k - 1}{k(k^2 - 4)} \sin\left(\frac{\pi k}{L} x\right) =$$

הנכנס

$$= \int \frac{128}{3\pi^2 L} \sum_{k=1}^{\infty} \frac{2}{(2k-1)(4 - (2k-1)^2)} \sin\left(\frac{\pi(2k-1)}{L} x\right)$$

הנכנס

$$\psi_E(x) = \frac{16}{\pi \sqrt{L}} \sum_{k=1}^{\infty} \frac{2}{(2k-1)(4 - (2k-1)^2)} \sin\left(\frac{\pi(2k-1)}{L} x\right)$$

(3) נרדף 2 פונקציות, הן נורמליות, הן זוגיות
 $\langle \psi_1 | \psi \rangle^2$ נחשב: ψ_1 נורמלית, ψ זוגית, ψ_1 אי-זוגית
 נרדף: $\psi_1 \sim \sin$

$$\langle \psi_1 | \psi \rangle = \int_{-\infty}^{\infty} \psi_1^*(x) \psi(x) dx = \int_0^L \sqrt{\frac{2}{L}} \sin\left(\frac{\pi}{L} x\right) \int \frac{8}{3L} \sin^2\left(\pi \frac{(L-x)}{L}\right) dx$$

הנכנס $\psi(x)$ הנכנס

$$= \frac{4}{L\sqrt{3}} \int_0^L \sin\left(\frac{\pi}{L} x\right) \sin^2\left(\pi \frac{(L-x)}{L}\right) dx =$$

$$\stackrel{\substack{= \\ \downarrow \\ \text{norm} \\ \text{onto}}}{=} \frac{4}{L\sqrt{3}} \int_0^L \sin^3\left(\frac{\pi x}{L}\right) dx =$$

$$= \frac{4}{L\sqrt{3}} \int_0^L \left(\sin\left(\frac{\pi x}{L}\right) - \sin\left(\frac{\pi x}{L}\right) \cos^2\left(\frac{\pi x}{L}\right) \right) dx$$

$$= \frac{4}{L\sqrt{3}} \left(-\frac{L}{\pi} \cos\left(\frac{\pi x}{L}\right) + \frac{1}{3} \frac{L}{\pi} \cos^3\left(\frac{\pi x}{L}\right) \right) \Bigg|_0^L =$$

$$= \frac{4}{\pi\sqrt{3}} \left[(-1) + \frac{(-1)^3}{3} - \left(-1 + \frac{1}{3} \right) \right] =$$

$$= \frac{4}{\pi\sqrt{3}} \left[\left(1 - \frac{1}{3} \right) - \left(-1 + \frac{1}{3} \right) \right] = \frac{4}{\sqrt{3}} \frac{1}{\pi} \frac{4}{3} = \frac{16}{3\sqrt{3}\pi}$$

psi

$$P_{1\psi}(\varphi_1) = |\langle \varphi_1 | \psi \rangle|^2 = \frac{256}{27\pi^2} \approx \boxed{0.961}$$

(74

מחשבים את
הערך הממוצע של
המיקום

$$\langle H \rangle = \langle \psi | H | \psi \rangle = \int_{-\infty}^{\infty} \langle \psi | x \rangle \langle x | H | \psi \rangle dx =$$

הערך הממוצע של המיקום

$$= \int_0^L \psi^*(x) \langle x | H | \psi \rangle dx =$$

הערך הממוצע של
המיקום

$$\int_0^L \sqrt{\frac{2}{3L}} \sin^2\left(\pi \frac{L-x}{L}\right) \left[-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \sqrt{\frac{2}{3L}} \sin^2\left(\pi \frac{L-x}{L}\right) \right] dx =$$

$$= \frac{8}{3L} \int_0^L dx \sin^2\left(\pi \frac{L-x}{L}\right) \left[-\frac{\hbar^2}{2m} \frac{2\pi^2}{L^2} \cos\left(\frac{2\pi(L-x)}{L}\right) \right]$$

$$= -\frac{8\hbar^2\pi^2}{3mL^3} \int_0^L \sin^2\left(\pi \frac{L-x}{L}\right) \cos\left(\frac{2\pi(L-x)}{L}\right) dx$$

$$= -\frac{8\hbar^2\pi^2}{3mL^3} \left(-\frac{L}{4}\right) = \boxed{\frac{2\hbar^2\pi^2}{3mL^2}}$$

... וואפן
wolfram-n

נכנסים למסלול

(4)

המשפט

$$\langle H \rangle = \sum_{n=1}^{\infty} E_n |\langle \psi_n | \psi \rangle|^2 = \sum_{n=1}^{\infty} E_n \left| \int_{-\infty}^{\infty} \varphi_n^*(x) \psi(x) dx \right|^2 =$$

$$= \sum_{n=1}^{\infty} \frac{16}{3L^2} \left(\frac{\pi n \hbar}{L \sqrt{2m}} \right)^2 \left(\frac{2L (\cos(\pi n) - 1)}{\pi n (n^2 - 4)} \right)^2$$

המשפט

$$= \frac{32 \hbar^2}{3mL^2} \sum_{n \text{ odd}} \frac{4}{(n^2 - 4)^2} = \frac{128 \hbar^2}{3mL^2} \sum_{n \text{ odd}} \frac{1}{(n^2 - 4)^2} =$$

$$= \frac{128 \hbar^2}{3mL^2} \frac{\pi^2}{64} =$$

$$\frac{2 \pi^2 \hbar^2}{3mL^2}$$

המשפט

(KS)

~ פנימי ~

$$E^{-2} = \langle H^2 \rangle = \langle \psi | H H | \psi \rangle = \langle \psi | H^\dagger H | \psi \rangle =$$

$$= (\langle \psi | H^\dagger) (H | \psi \rangle) = \| H \psi \|^2 =$$

~ ממוצע ~ נורמה וינטגרה פנימית ~

$\|v\|^2 = \langle v | v \rangle$

$$= \int_{-\infty}^{\infty} dx \left| \langle x | \frac{p^2}{2m} | \psi \rangle \right|^2 = \int_{-\infty}^{\infty} dx \left| -\frac{\hbar^2}{2m} \psi''(x) \right|^2$$

$H \sim p^2$
 (קורטו) $\sim \hbar^2$
 (מיקרו) $\sim m$
 (מיקרו) $\sim m$

$$= \frac{\hbar^4}{4m^2} \int_0^L dx \frac{8}{3L} \left(\partial_x^2 \left[\sin^2 \left(\pi \frac{L-x}{L} \right) \right] \right)^2 =$$

$$= \frac{8\hbar^4}{12Lm^2} \int_0^L \left(\partial_x \left(2 \sin \left(\pi \frac{L-x}{L} \right) \cos \left(\pi \frac{L-x}{L} \right) \right) \right)^2 dx =$$

$$= \frac{8\hbar^4 \pi^2}{12L^3 m^2} \int_0^L \partial_x \left(-\frac{2\pi}{L} \cos \left(2\pi \frac{L-x}{L} \right) \right)^2 dx =$$

~ פנימי ~

$$= \frac{8 \hbar^4 \pi^2}{12 L^3 m^2} \frac{4\pi^2}{L^2} \int_0^L \left(\frac{1}{2} + \frac{1}{2} \cos \left(4\pi \frac{L-x}{L} \right) \right) dx =$$

$$= \frac{8 \hbar^4 \pi^2}{12 L^3 m^2} \frac{4\pi^2}{L^2} \cdot \frac{L}{2} = \frac{4 \cdot 8 \hbar^4 \pi^4}{12 \cdot 2 \cdot L^4 m^2} = \boxed{\frac{4 \hbar^4 \pi^4}{3 L^4 m^2}}$$

يس:

$$(\Delta E)^2 = \overline{E^2} - (\overline{E})^2 = \frac{4 \hbar^4 \pi^4}{3 L^4 m^2} - \left(\frac{2 \pi^2 \hbar^2}{3 m L^2} \right)^2 =$$

$$= \frac{8 \hbar^4 \pi^4}{9 m^2 L^4}$$

$\Rightarrow$

$$\boxed{\Delta E = \frac{2\sqrt{2} \hbar^2 \pi^2}{3 m L^2}}$$

(25)

$$\bar{E}^2 = \langle H^2 \rangle = \langle \psi | H^2 | \psi \rangle =$$

$$= \int_{-\infty}^{\infty} \langle \psi | H | x \rangle \langle x | H | \psi \rangle dx =$$

הפעולה הפנימית של הפונקציה הנורמלית

$$\langle x | H | \psi \rangle = \langle x | \left(-\frac{\hbar^2}{2m} \partial_x^2 \psi(x) \right) | \psi \rangle$$

הנורמליזציה

$$= \int_0^L \left(-\frac{\hbar^2}{2m} \partial_x^2 \left[\sqrt{\frac{8}{3L}} \sin^2 \left(\pi \frac{L-x}{L} \right) \right] \right)^2 dx =$$

$$= \frac{\hbar^4}{4m^2} \frac{8}{3L} \int_0^L \left(\partial_x^2 \sin^2 \left(\pi \frac{L-x}{L} \right) \right)^2 dx =$$

$$= \frac{\hbar^4}{4m^2} \frac{8}{3L} \int_0^L \left(\frac{2\pi^2 \cos \left(\frac{2\pi(L-x)}{L} \right)}{L^2} \right)^2 dx =$$

$$= \frac{32 \hbar^4 \pi^4}{12 L^5 m^2} \int_0^L \cos^2 \left(\frac{2\pi(L-x)}{L} \right) dx =$$

$$= \frac{32 \hbar^4 \pi^4}{12 L^5 m^2} \int_0^L \left(\frac{1}{2} + \frac{1}{2} \cos \left(\frac{4\pi(L-x)}{L} \right) \right) dx =$$

$$= \frac{32 \hbar^4 \pi^4}{12 L^5 m^2} \cdot \frac{L}{2} = \boxed{\frac{4 \hbar^4 \pi^4}{3 L^4 m^2}}$$

پس،

$$(\Delta E)^2 = \bar{E}^2 - (\bar{E})^2 = \frac{4 \hbar^4 \pi^4}{3 L^4 m^2} - \left(\frac{2 \pi^2 \hbar^2}{3 m L^2} \right)^2 =$$

$$= \frac{8 \hbar^4 \pi^4}{9 m^2 L^4} \Rightarrow \boxed{\Delta E = \frac{2\sqrt{2} \hbar^2 \pi^2}{3 m L^2}}$$

25) $\int e^{-3x} dx$ סכום הסדרות!

$$\bar{E}^2 = \langle H^2 \rangle = \sum_{n=1}^{\infty} E_n^2 |\langle \varphi_n | \Psi \rangle|^2 =$$

↓
הנורמה של Ψ

$$= \sum_{n=1}^{\infty} E_n^2 \left| \int_{-\infty}^{\infty} \varphi_n^*(x) \Psi(x) dx \right|^2 =$$

הנורמה של Ψ

$$= \sum_{n=1}^{\infty} \frac{16}{3L^2} \frac{\pi^4 n^4 \hbar^4}{4m^2 L^4} \left(\frac{2L(\cos(\pi n) - 1)}{\pi n(n^2 - 4)} \right)^2 =$$

הנורמה של Ψ

$$= \frac{64\pi^2 \hbar^4}{3m^2 L^4} \sum_{n \in \text{odd}} \frac{n^2}{(n^2 - 4)^2} = \frac{64\pi^2 \hbar^4}{3m^2 L^4} \cdot \frac{\pi^2}{16} = \boxed{\frac{4\pi^4 \hbar^4}{3m^2 L^4}}$$

הנורמה של Ψ

$$(\Delta E)^2 = \bar{E}^2 - (\bar{E})^2 = \frac{4 \hbar^4 \pi^4}{3L^4 m^2} - \left(\frac{2\pi^2 \hbar^2}{3mL^2} \right)^2 =$$

$$= \frac{8 \hbar^4 \pi^4}{9m^2 L^4} \Rightarrow$$

$$\boxed{\Delta E = \frac{2\sqrt{2} \hbar^2 \pi^2}{3mL^2}}$$

3: סעיף

הערה 2: קורס נא - הכוללים הערה

ההארה הערה קבוצה נוספת איננו:

$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi}{L}x\right), \quad E_n = \frac{\pi^2 \hbar^2 n^2}{2mL^2}$$

1) נניח כי הנהג - הנהג קבוצה:

הנהג - הנהג

$$\sin \alpha \cos \beta = \frac{1}{2} [\sin(\alpha + \beta) + \sin(\alpha - \beta)]$$

$$\psi(x,0) = A_1 \sin\left(\frac{3\pi x}{L}\right) \cos\left(\frac{\pi x}{L}\right) = \frac{A_1}{2} \left[\sin\left(\frac{4\pi x}{L}\right) + \sin\left(\frac{2\pi x}{L}\right) \right]$$

הנהג - הנהג A_1 - הנהג

$$1 = \langle \psi_0 | \psi_0 \rangle = \int_0^L \frac{1}{4} |A_1|^2 \left[\sin\left(\frac{4\pi x}{L}\right) + \sin\left(\frac{2\pi x}{L}\right) \right]^2 dx$$

$$= \frac{|A_1|^2}{4} \int_0^L dx \left(1 - \frac{1}{2} \cos\left(\frac{8\pi x}{L}\right) - \frac{1}{2} \cos\left(\frac{4\pi x}{L}\right) + \sin\left(\frac{4\pi x}{L}\right) \sin\left(\frac{2\pi x}{L}\right) \right) =$$

האם זה נכון?!

$$= \frac{|A_1|^2}{4} L + \frac{|A_1|^2}{4} \int_0^L dx \sin\left(\frac{4\pi x}{L}\right) \sin\left(\frac{2\pi x}{L}\right) = \frac{|A_1|^2}{4} L$$

↓

$$A_1 = \frac{2}{\sqrt{L}}$$

נניח $H = H_0 + V$ ונניח $V = 0$

אז $V = 0$

$$U(t) = e^{-i \frac{E_1}{\hbar} t}$$

נניח $\psi(x, t) = A_1 \sin\left(\frac{2\pi x}{L}\right) e^{-i \frac{E_1}{\hbar} t}$ ונניח $A_1 = \frac{2}{\sqrt{L}}$

$$\psi(x, t) = \frac{1}{\sqrt{L}} e^{-i \frac{8\pi^2 \hbar}{m L^2} t} \sin\left(\frac{4\pi x}{L}\right) + \frac{1}{\sqrt{L}} e^{-i \frac{2\pi^2 \hbar}{m L^2} t} \sin\left(\frac{2\pi x}{L}\right)$$

אם $P(E_1) = 1$ ו- $P(E_2) = 0$ אז $P(E_1) = 1$ ו- $P(E_2) = 0$

אם $P(E_1) = 1$ ו- $P(E_2) = 0$ אז $P(E_1) = 1$ ו- $P(E_2) = 0$

$$P(E_2) = |\langle \varphi_2 | \Psi \rangle|^2 = \left| \int_0^L \sqrt{\frac{2}{L}} \sin\left(\frac{2\pi}{L}x\right) \cdot \right.$$

$$\left. \cdot \left(\frac{1}{\sqrt{L}} e^{-i \frac{8\pi^2 \hbar t}{mL^2}} \sin\left(\frac{4\pi x}{L}\right) + \frac{1}{\sqrt{L}} e^{-i \frac{2\pi^2 \hbar t}{mL^2}} \sin\left(\frac{2\pi x}{L}\right) \right) \right|^2 =$$

$$\stackrel{\text{Nichtlinear}}{=} \frac{1}{L} \left| \int_0^L dx \sin^2\left(\frac{2\pi x}{L}\right) \right|^2 = \frac{1}{L} \left| \int_0^L dx \left(\frac{1}{2} - \frac{1}{2} \cos\left(\frac{4\pi x}{L}\right) \right) \right|^2 =$$

$$= \frac{2}{L^2} \left| \frac{L}{2} \right|^2 = \frac{2}{4} = \boxed{\frac{1}{2}}$$

$$P(E_4) = |\langle \varphi_4 | \Psi \rangle|^2 = \left| \int_0^L \sqrt{\frac{2}{L}} \sin\left(\frac{4\pi}{L}x\right) \cdot \right.$$

$$\left. \cdot \left(\frac{1}{\sqrt{L}} e^{-i \frac{8\pi^2 \hbar t}{mL^2}} \sin\left(\frac{4\pi x}{L}\right) + \frac{1}{\sqrt{L}} e^{-i \frac{2\pi^2 \hbar t}{mL^2}} \sin\left(\frac{2\pi x}{L}\right) \right) \right|^2 =$$

$$\stackrel{\text{Nichtlinear}}{=} \frac{1}{L} \left| \int_0^L dx \sin^2\left(\frac{4\pi x}{L}\right) \right|^2 = \frac{1}{L} \left| \int_0^L dx \left(\frac{1}{2} - \frac{1}{2} \cos\left(\frac{8\pi x}{L}\right) \right) \right|^2 =$$

$$= \frac{2}{L^2} \left| \frac{L}{2} \right|^2 = \frac{2}{4} = \boxed{\frac{1}{2}}$$

! 1-5 non-normal pic

$$\psi(x,0) = A_2 x^2 (x-L)^2 \quad : \text{norm} \quad (2)$$

: نرمال سازی کنیم A_2 را

$$1 = \langle \psi_0 | \psi_0 \rangle = |A_2|^2 \int_0^L x^4 (x-L)^4 dx =$$

$$= |A_2|^2 \int_0^L x^4 (x^2 - 2xL + L^2)^2 dx =$$

$$= |A_2|^2 \int_0^L dx x^4 (x^4 - 2x^3L + L^2x^2 - 2x^2L + 4xL^2 - 2xL^3 + L^4) =$$

$$= |A_2|^2 \int_0^L dx x^4 (x^4 - 4x^3L + 6x^2L^2 - 4xL^3 + L^4)$$

$$= |A_2|^2 \int_0^L dx (x^8 - 4x^7L + 6x^6L^2 - 4x^5L^3 + x^4L^4)$$

$$= |A_2|^2 \left(\frac{1}{9} L^9 - \frac{4}{8} L^8 L + \frac{6}{7} L^7 L^2 - \frac{4}{6} L^6 L^3 + \frac{1}{5} L^5 L^4 \right) = |A_2|^2 \frac{L^9}{630} \Rightarrow A_2 = \sqrt{\frac{630}{L^9}}$$

↓
נורמליזציה

הנורמליזציה של ψ נכונה?

$|\varphi_n\rangle$ סדרת הנורמליזציה

$$\Psi(x,t) = \langle x | \psi(t) | \psi \rangle = \sum_{n=1}^{\infty} \langle x | e^{-i \frac{E_n}{\hbar} t} | \varphi_n \rangle \langle \varphi_n | \psi_0 \rangle$$

הנורמליזציה

$$\langle \varphi_n | \psi_0 \rangle = \int_{-\infty}^{\infty} \varphi_n^*(x) \psi(x) dx =$$

$$= \int_0^L \sqrt{\frac{2}{L}} \sin\left(\frac{\pi n}{L} x\right) \sqrt{\frac{630}{L^9}} x^2 (x-L)^2 dx =$$

$$= \frac{6\sqrt{35}}{L^5} \int_0^L \sin\left(\frac{n\pi}{L} x\right) x^2 (x-L)^2 dx =$$

$$\frac{6\sqrt{35}}{L^5} \frac{2L^5 (L^2 \pi^2 n^2 - 12) (\cos(\pi n) - 1) - 6\pi n \sin(\pi n)}{\pi^5 n^5}$$

↓
wolfram

$$= \frac{12\sqrt{35}}{\pi^5 h^5} \left((\pi^2 h^2 - 12) (\cos(\pi h) - 1) - 6\pi h \overset{0}{\sin(\pi h)} \right)$$

ואכן: h מסומן כאינסוף, $\psi(x,t)$ סופרפוזיציה

מכאן ψ_h נקראת פונקציית גל, כלומר E_h נקראת אנרגיה (הקוונטום)

$$\psi(x,t) = \sum_{h \in \text{odd}} \frac{24\sqrt{35}}{\pi^5 h^5 L} (12 - \pi^2 h^2) \sin\left(\frac{\pi h}{L} x\right) e^{-i \frac{\pi^2 h^2 \hbar}{2mL^2} t}$$

אנרגיה E_h וקוונטום $\hbar$

$$P(E_h) = |\langle \psi_h | \psi \rangle|^2 = \left| \frac{12\sqrt{35}}{\pi^5 h^5} (\pi^2 h^2 - 12) (\cos(\pi h) - 1) \right|^2$$

ψ_h נקראת פונקציית גל

כלומר:

$$P(E_h) = \begin{cases} 0 & , h \text{ זוגי} \\ \frac{20160}{\pi^{10} h^{10}} (\pi^2 h^2 - 12)^2 & , h \text{ אי-זוגי} \end{cases}$$

4 75Kc

$$\langle p | \psi_0 \rangle = \psi_0(p)$$

①

$$\psi(x, t) = \langle x | U(t) | \psi \rangle = \int_{-\infty}^{\infty} dp \langle x | p \rangle \langle p | U(t) | \psi \rangle$$

$$= \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} dp e^{\frac{ipx}{\hbar}} \langle p | e^{-\frac{ip^2 t}{2m\hbar}} | \psi_0 \rangle =$$

$$\langle x | p \rangle, U(t) = e^{-\frac{ip^2 t}{2m\hbar}}$$

$$= \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} dp e^{\frac{ipx}{\hbar}} e^{-\frac{ip^2 t}{2m\hbar}} \psi_0(p) =$$

$$\langle p | \psi_0 \rangle = \psi_0(p)$$

$$\frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} dp \psi_0(p) e^{\frac{i}{\hbar} \left(px - \frac{p^2}{2m} t \right)}$$

2

הערך $\psi(0,0)$ של פונקציית הגל:

$$\psi(0,0) = 1.09 \times 10^{-23} - 3.67 \times 10^{-23} i$$

קבוע הנורמליזציה N :

$$N = 7.1314 \times 10^{16}$$

עכשיו נחלק את המכונה ל-3 חלקים

1) חלק קטן: part2_6.gif

5) חלק קטן: part2_7.gif

3

האנמליזציה של המכונה

הקבוע:

part3_1.gif

part3_2.gif

part3_3.gif

החלקים 3-N של המכונה

החלקים 3-N של המכונה:

yovel\QP1\hw7_4.py

```

1
2 import numpy as np
3 import matplotlib.pyplot as plt
4 from matplotlib.animation import FuncAnimation
5
6 # ***** (2.1) *****
7
8 def integrate_riemann(f):
9     dy = 20.0 / (len(f) - 1)
10    return np.sum(f) * dy
11
12 # ***** (2.2) *****
13
14 print("\nThe first steps of the exercise")
15 print("*****")
16
17 h_bar = 1.054571817e-34
18
19 def phi_0(p):
20     N = 1
21     L = 1.0 # [cm]
22
23     p0 = h_bar / L
24     delta = h_bar / L
25     x0 = -5 * L
26
27     gaussian = np.exp(-((p - p0) ** 2) / (2 * (delta ** 2)))
28
29     phase = np.exp(-1j * p * x0 / h_bar)
30
31     return N * gaussian * phase
32
33 def phi(x, t):
34     L = 1.0 # [cm]
35     p0 = h_bar / L
36     m = h_bar / L
37
38     y = np.linspace(-10, 10, 2000)
39
40     p = y * p0
41
42     free = (p ** 2) / (2 * m)
43     exponent = (1j / h_bar) * (p * x - free * t)
44
45     integrand = phi_0(p) * np.exp(exponent)
46
47     integral_y = integrate_riemann(integrand)
48
49     integral_p = integral_y * p0
50
51     normalization_factor = 1 / np.sqrt(2 * np.pi * h_bar)

```

```

52
53     return normalization_factor * integral_p
54
55 print(f" $\psi(0, 0) = \{\text{phi}(0, 0)\}$ ")
56
57 # ***** (2.3) *****
58
59 x = np.arange(-10, 10.1, 0.1)
60
61 psi_unnormalized = np.array([phi(x, 0) for x in x])
62
63 # ***** (2.4) *****
64
65 probability_density = np.abs(psi_unnormalized) ** 2
66
67 integral_x = integrate_riemann(probability_density)
68
69 N = 1 / np.sqrt(integral_x)
70
71 print(f"N = {N}")
72
73 # ***** (2.5) *****
74
75 x_vec = np.arange(-10, 10.1, 0.1)
76 t_vec = np.arange(0, 10.1, 0.1)
77
78 # empty matrix, rows will represent time steps, columns will represent spatial points.
79 psi_xt = np.zeros((len(t_vec), len(x_vec)), dtype=complex)
80
81 for i, t in enumerate(t_vec):
82     for j, x in enumerate(x_vec):
83         psi_xt[i, j] = N * phi(x, t)
84
85 # ***** (2.6) *****
86
87 prob_density = np.abs(psi_xt)**2
88
89 fig, ax = plt.subplots(figsize=(8, 5))
90
91 ax.set_xlim(-10, 10)
92 ax.set_ylim(0, np.max(prob_density) * 1.1)
93
94 ax.set_title('Time Evolution of Probability Density  $|\psi(x,t)|^2$ ')
95 ax.set_xlabel('Position (x)')
96 ax.set_ylabel('Probability Density')
97 ax.grid(True)
98
99 line, = ax.plot([], [], lw=2, color='blue')
100
101 # required for this graphic library
102 def init():
103     line.set_data([], [])
104     return line,
105

```

```

106 # required for this graphic librery
107 def animate(i):
108     line.set_data(x_vec, prob_density[i, :])
109     ax.set_title(f'Time Evolution ($t = {t_vec[i]:.1f}$)')
110     return line,
111
112 # main animation engine
113 anim = FuncAnimation(fig, animate, init_func=init,
114                     frames=len(t_vec), interval=100, blit=True)
115
116 # save to GIF
117 anim.save('part2_6.gif', writer='pillow', fps=10)
118
119 # generate the animation
120 plt.show()
121
122 # ***** (2.7) *****
123
124 print("\npart 2.7:")
125 print("*****")
126
127 h_bar = 1.054571817e-34
128
129 # The Fourier transform of the given function which I assume should be:  $\psi(x, 0)$ 
130 def phi_0_zayin(p):
131     N = 1
132     L = 1.0
133
134     p0 = h_bar / L
135     delta = L
136     x0 = -5.0 * L
137
138     lorentz = np.exp(- (np.abs(p - p0) * delta) / h_bar)
139
140     phase = np.exp(-1j * p * x0 / h_bar)
141
142     return N * lorentz * phase
143
144 # The rest of the code is the same as the previous parts of the question...
145
146 def phi(x, t):
147     L = 1.0 # [cm]
148     p0 = h_bar / L
149     m = h_bar / L
150
151     y = np.linspace(-10, 10, 2000)
152
153     p = y * p0
154
155     free = (p ** 2) / (2 * m)
156     exponent = (1j / h_bar) * (p * x - free * t)
157
158     integrand = phi_0_zayin(p) * np.exp(exponent)
159

```

```

160     integral_y = integrate_riemann(integrand)
161
162     integral_p = integral_y * p0
163
164     normalization_factor = 1 / np.sqrt(2 * np.pi * h_bar)
165
166     return normalization_factor * integral_p
167
168     print(f" $\psi(0, 0) = \{\phi(0, 0)\}$ ")
169
170     x = np.arange(-10, 10.1, 0.1)
171
172     psi_unnormalized = np.array([phi(x, 0) for x in x])
173
174     probability_density = np.abs(psi_unnormalized) ** 2
175
176     integral_x = integrate_riemann(probability_density)
177
178     N = 1 / np.sqrt(integral_x)
179
180     print(f"N = {N}")
181
182     x_vec = np.arange(-10, 10.1, 0.1)
183     t_vec = np.arange(0, 10.1, 0.1)
184
185     # empty matrix, rows will represent time steps, columns will represent spatial points.
186     psi_xt = np.zeros((len(t_vec), len(x_vec)), dtype=complex)
187
188     for i, t in enumerate(t_vec):
189         for j, x in enumerate(x_vec):
190             psi_xt[i, j] = N * phi(x, t)
191
192     prob_density = np.abs(psi_xt)**2
193
194     fig, ax = plt.subplots(figsize=(8, 5))
195
196     ax.set_xlim(-10, 10)
197     ax.set_ylim(0, np.max(prob_density) * 1.1)
198
199     ax.set_title('Time Evolution of Probability Density  $|\psi(x,t)|^2$ ')
200     ax.set_xlabel('Position (x)')
201     ax.set_ylabel('Probability Density')
202     ax.grid(True)
203
204     line, = ax.plot([], [], lw=2, color='blue')
205
206     # required for this graphic library
207     def init():
208         line.set_data([], [])
209         return line,
210
211     # required for this graphic library
212     def animate(i):
213         line.set_data(x_vec, prob_density[i, :])

```



```

214     ax.set_title(f'Time Evolution ($t = \{t\_vec[i]:.1f\}$)')
215     return line,
216
217 # main animation engine
218 anim = FuncAnimation(fig, animate, init_func=init,
219                     frames=len(t_vec), interval=100, blit=True)
220
221 # save to GIF
222 anim.save('part2_7.gif', writer='pillow', fps=10)
223
224 # generate the animation
225 plt.show()
226
227 # ***** part 3 *****
228
229 print("\npart 3:")
230 print("*****")
231
232 h_bar = 1.0
233 m = 1.0
234 V0 = 5.0
235 x0 = -15.0
236 Delta_q = 0.5
237
238 # (E = V0)
239 q_critical = np.sqrt(2 * m * V0)
240
241 # Here you decide which section to run!
242 # =====
243 # a) (E < V0): q0 = 2.0
244 # b) (E > V0): q0 = 4.0
245 # c) (E ~ V0): q0 = q_critical
246 # =====
247
248 part_number = 1
249 parts = [2.0, 4.0, q_critical]
250
251 for part in parts:
252     q0 = part
253
254     def phi_step(x, t):
255         q_values = np.linspace(q0 - 6*Delta_q, q0 + 6*Delta_q, 1000)
256         dq = q_values[1] - q_values[0]
257
258         phi_0_q = np.exp(-((q_values - q0)**2) / (2 * Delta_q**2)) * np.exp(-1j * q_values
259 * x0 / h_bar)
260
261         E_q = (q_values**2) / (2 * m)
262
263         time_phase = np.exp(-1j * E_q * t / h_bar)
264
265         integral_sum = 0
266
267         for i, q in enumerate(q_values):

```

```

267     k_prime = np.sqrt(q**2 - 2 * m * V0 + 0j)
268
269     R = (q - k_prime) / (q + k_prime)
270     T = 2 * q / (q + k_prime)
271
272     if x < 0:
273         psi_xq = np.exp(1j * q * x / h_bar) + R * np.exp(-1j * q * x / h_bar)
274     else:
275         psi_xq = T * np.exp(1j * k_prime * x / h_bar)
276
277     integral_sum += phi_0_q[i] * psi_xq * time_phase[i] * dq
278
279     return integral_sum
280
281     # From here it's similar to the previous code... just creating the animation
282
283     print(f"Critical Momentum for E=V0 is: {q_critical:.2f}")
284     print("Calculating space-time grid. This will take a moment...")
285
286     x_values = np.arange(-30, 20, 0.2)
287     t_values = np.arange(0, 15, 0.1)
288
289     psi_xt = np.zeros((len(t_values), len(x_values)), dtype=complex)
290
291     for i, t in enumerate(t_values):
292         for j, x in enumerate(x_values):
293             psi_xt[i, j] = phi_step(x, t)
294
295     prob_density = np.abs(psi_xt)**2
296     prob_density = prob_density / np.max(prob_density)
297
298     fig, ax = plt.subplots(figsize=(10, 5))
299     ax.set_xlim(-30, 20)
300     ax.set_ylim(0, 1.1)
301
302     # We will see the potential step as a red line
303     ax.plot([0, 0], [0, 1.1], color='red', linestyle='--', alpha=0.5, label='Potential
Step (x=0)')
304     ax.fill_between([0, 20], 0, 1.1, color='red', alpha=0.05)
305
306     line, = ax.plot([], [], lw=2, color='blue', label='$|\psi(x,t)|^2$')
307     ax.legend(loc='upper right')
308     ax.set_xlabel('Position (x)')
309     ax.set_ylabel('Normalized Probability')
310
311     def init():
312         line.set_data([], [])
313         return line,
314
315     def animate(i):
316         line.set_data(x_values, prob_density[i, :])
317         ax.set_title(f'Wavepacket Scattering ($E_0$ proxy $q_0={q0}$) | Time:
{t_values[i]:.1f}')
318         return line,

```

```
319
320     anim = FuncAnimation(fig, animate, init_func=init,
321                           frames=len(t_values), interval=100, blit=True)
322
323     # save to GIF
324     anim.save(f'part3_{part_number}.gif', writer='pillow', fps=10)
325     part_number += 1
326
327     plt.show()
```